SEPARATED IMPLIES QUASI-SEPARATED, DM IMPLIES QUASI-DM, AND REPRESENTABLE IMPLIES DM

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the Stacks Project keeps one single shared preamble for every chapter; this card takes it directly from the staged sources/stacks-algebraic-stacks-morphisms/preamble.tex as three verbatim spans (lines 8-12, 14-18 and 154-217), so no derived file was written and nothing was added or redefined. The wrapper class and the base packages are supplied by the assembler.

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Editorial note. Selection provenance: this card was opened under the file-based control contract (CONTROL.md), with the mechanical pre-card gate run on its statement and proof interval before the card was written.

Allowed standard prerequisites (editorial).

- The definition of a proper morphism of algebraic spaces. Invoked inside the counted proof (source lines 692-694) as a cross-chapter pointer. Gives that a proper morphism of algebraic spaces is quasi-compact and quasi-separated.
- The statement that an unramified morphism of algebraic spaces is locally quasi-finite. Invoked inside the counted proof (source lines 695-697) as a cross-chapter pointer. Gives local quasi-finiteness from unramifiedness.
- The characterization of representability of a morphism by its diagonal. Invoked at the end of the counted proof (source lines 698-699). Deduces the third implication.

Source reference. Original lemma lemma-properties-diagonal-representable: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 519–519. Navigation pointer only: the counted text refers to the Stacks item lemma-properties-diagonal-representable, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference spaces-morphisms-definition-proper: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 694–694. Navigation pointer only: the counted text refers to the Stacks item spaces-morphisms-definition-proper, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Date: stacks-project:stacks-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference spaces-morphisms-lemma-unramified-quasi-finite: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 698–698. Navigation pointer only: the counted text refers to the Stacks item spaces-morphisms-lemma-unramified-quasi-finite, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let $f : \mathcal{X} \to \mathcal{Y}$ be a morphism of algebraic stacks.*

(1) *If f is separated, then f is quasi-separated.*
(2) *If f is DM, then f is quasi-DM.*
(3) *If f is representable by algebraic spaces, then f is DM.*

Proof. To see (1) note that a proper morphism of algebraic spaces is quasi-compact and quasi-separated, see Morphisms of Spaces, Definition spaces-morphisms-definition-proper. To see (2) note that an unramified morphism of algebraic spaces is locally quasi-finite, see Morphisms of Spaces, Lemma spaces-morphisms-lemma-unramified-quasi-finite. Finally (3) follows from Lemma lemma-properties-diagonal-representable. □